Lecture 13 (October 28th)

Recall. Let $f(t) \in K[t]$ and E be the splitting field for $f(t)$. We have seen (1) $[E : K]$ and (2) the number of K-embeddings $E \to \bar{K}$. Today's goal is to study (2) and separability.

Chapter 14.4 Separable Extensions

Definition. (14.9) Let K be a field. A polynomial $f(t) \in K[t]$ is separable if its factorisation over $\bar{K}$ has no multiple roots. Let F/K be a algebraic extension. An element $\alpha \in F$ is separable over K if the minimal polynomial $\mathrm{irr}(\alpha, K)$ over K is separable. We say that F/K is separable if every $\alpha \in F$ is separable over K.

Proposition. Let $f(t) \in K[t]$ be a nonzero polynomial. Then, $f(t)$ is separable over K if $f(t)$ has no multiple roots in $\bar{K}$. The following are equivalent:

(i) $f(t)$ is separable over K

(ii) $f(t)$ and $f'(t)$ have no common root in $\bar{K}$

(iii) $\gcd(f(t), f'(t)) = 1$

Proof. The proof is yours! □

Proposition. (14.23) Let K be a field of characteristic zero and let $f(t) \in K[t]$ be a irreducible polynomial. Then, $f(t)$ is separable. Moreover, algebraic extensions of fields of characteristic zero are separable.

Proof. Set $f(t) = a_n t^n + \ldots + a_1 t + a_0$ so that $f'(t) = n a_n t^{n-1} + \ldots + a_1$. Note that $\deg(f') < \deg(f)$ since $\mathrm{char}(K) = 0$. Suppose that $\alpha \in \bar{K}$ is a common root of $f(t)$ and $f'(t)$. Then $\mathrm{irr}(\alpha, K) | f'(t), f(t)$. Since $f(t)$ is irreducible over K, $\deg\big(\mathrm{irr}(\alpha, K)\big) = \deg\big(f(t)\big)$. This contradicts to the observation that $\mathrm{irr}(\alpha, K) | f'(t)$ because $\deg(f') < \deg(f)$. □

Proposition. (14.24) (Primative element theorem) Every finite separable extension is simple.

Proof. The proof is ours and is technical. □

Corollary. (14.25) Let F/K be a finite separable extension. Then there are exactly $[F : K]$ number of K-embeddings $F \to \bar{K}$.

Proof. By the primative element theorem, $F = K(\alpha)$ for some $\alpha \in F$. Since α is separable over K, $\operatorname{irr}(\alpha, K)$ has no multiple roots in $\bar{K}$. Applying proposition (14.3), the number of K-embeddings $F = K(\alpha) \hookrightarrow \bar{K}$ equals the number of distinct roots in $\bar{K}$ of $\operatorname{irr}(\alpha, K)$ which is equal to

$$\deg\big(\operatorname{irr}(\alpha, K)\big) = [K(\alpha) : K]$$

□

Remark. Let $\alpha \in F$ be an algebraic element over K. Then the number of K-embeddings $K(\alpha) \hookrightarrow \bar{K}$ is the number of distinct zeros of $\operatorname{irr}(\alpha, K)$ in $\bar{K}$, and both are less or equal to $\deg\big(\operatorname{irr}(\alpha, K)\big) = [K(\alpha), K]$. According to corollary (14.25), the euqality holds if and only if α is separable over K.

Definition. (14.26) The separability degree of an algebraic extension F/K denoted $[F : K]_s$, is the number of K-embeddings $F \hookrightarrow \bar{K}$.

Lecture 14 (October 30th)

Remark. Last class, we learned about separable extensions, and today we will be learning about separability degree.

Definition. (14.9) We called a $\alpha \in F$ separable if $\operatorname{irr}(\alpha, K)$ has no repeated roots in $\bar{K}$.

Proposition. (14.23) If $\operatorname{char}(K) = 0$, every irreducible polynomial in $K[x]$ is separable.

Proposition. (14.24) (Prime element theorem) Finite separable extensions are simple.

Corollary. (14.25) For a finite separable extension F/K, the number of K-embeddings $F \to \langle K$ is $[F : K]$. In fact, for an algebraic $\alpha \in F$ over K,

$$|\{F \xrightarrow{\sigma} \bar{K} \mid \sigma|_K = 1_K\}| = |\{\beta \in \bar{K} \mid (\operatorname{irr}(\alpha, K))(\beta) = 0\}| \leq \deg(\operatorname{irr}(\alpha, K)) = [K(\alpha), K]$$

Definition. (14.26) Let F/K be an algebraic extension. Then

$$[F : K]_s = |\{\sigma : F \to \bar{K} \mid \sigma|_K = 1_K\}$$

where $[F : K]_s$ denotes the separability degree of F/K.

Remark. Definition (14.26) is independent of the choice of the algebraic closure of $\bar{K}$.

Proposition. (14.28) If F/E and E/K are algebraic extensions then

$$[F : K]_s = [F : E]_s [E : K]_s$$

Proof. Proof will be further inquired in the supplementary material. $\square$

Theorem. Let F/K be a finite extension. Then F/K is separable if and only if $[F:K]_s = [F:K]$.

Remark. In the case where $F = K(\alpha)$, we already have observed the result.

Remark. For a finite extension F/K,

$$[F:K]_s \leq [F:K]$$

Proof. (if) Let $\alpha \in F$. Consider the tower $F/K(\alpha)/K$. Note that

$$\begin{aligned}[F:K] =& [F:K(\alpha)][K(\alpha):K] \\ [F:K]_s =& [F:K(\alpha)]_s[K(\alpha):K]_s\end{aligned}$$

and $[F:K]$ implies that, through the remark, $[F:K(\alpha)] = [F:K(\alpha)]_s$ and $[K(\alpha):K] = [K(\alpha):K]_s$. That is, α is separable over K.

(only if) Ours! $\square$

Corollary. Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be elements of $\bar{K}$. Then $\alpha_1, \ldots, \alpha_n$ are separable over K if and only if $K(\alpha_1, \ldots, \alpha_n)$ is separable over K.

Corollary. Let F/E and E/K be finite extensions. F/K is separable if and only if F/E and E/K are separable.

Proof. To prove this, go back to an analogous statement for algebraic extensions. $\square$

Corollary. Let F/K be an algebraic extension then the set of separable elements of F over K forms a subfield of F.

Lecture 15 (November 4th)

Remark. Last time, we saw various properties of separable extensions. Today, we will learn about normal extensions.

Proposition. (14.23) If $\mathrm{char}(K) = 0$, then every algebraic extension of K is separable.

Proposition. (14.24) (Primitive element theorem) Every finite separable extension is simple.

Definition. (14.26) Let F/K be an algebraic extension. $[F : K]_s$ denotes the separability degree of F/K which is equal to the number of K-embeddings from $F \to \bar{K}$.

Proposition. (14.28) Let $F/E/K$ be algebraic extensions.

$$[F : K]_s = [F : E]_s[E : K]_s$$

Corollary. If F/K is finite, $[F : K]_s < \infty$ (and $[F : K]_s$ divides $[F : K]$).

Chapter 14.3 Normal Extensions

Definition. (14.15) An algebraic extension F/K is normal if for every $\alpha \in F$, the minimal polynomial $\operatorname{irr}(\alpha, K)$ splits in F.

Remark. F/K is normal if and only if the following condition holds: if F contains a zero of an irreducible polynomial $p(t) \in L[t]$, then $p(t)$ splits into linear factors over F.

Remark. (Motivation) Let $p(t) \in K[t]$ be irreducible. Say $p(t) = \prod_{i=1}^{n}(t - \lambda_i) \in \bar{K}[t]$. Consider a K-embedding $\sigma : E\bar{K}$ where $E = K(\lambda_1, \ldots, \lambda_n)$ is the splitting field of $p(t)$. We want to figure out the image $\sigma(E) \subset \bar{K}$. We claim that $\sigma(E) = E$. For this, observe that the image of λ_i under σ is α_j for some j. In other words, σ permutes $\{\lambda_1, \lambda_2, \ldots, \lambda_n\}$ ($\sigma : \{\lambda_1, \ldots, \lambda_n\} \to \{\lambda_1, \ldots, \lambda_n\}$). All in all, σ gives rise to a K-automorphism on E.

Theorem. Let F/K be an algebraic extension. Fix an algebraic closure $\bar{K}$ of K with $K \subset F \subset \bar{K}$. The following are equivalent:

(i) There exists a subset $S \subset K[t]$ such that F is a splitting field for S.

(ii) If $\sigma : F \to \bar{K}$ is a K-embedding, then $\sigma(F) = F$.

(iii) F/K is normal.

Proof. Assume that condition (ii) holds. To show that F is normal over K, let α be an element of F. Suppose we are given a zero β of $\operatorname{irr}(\alpha, K)$ in $\bar{K}$. We wish to show that $\beta \in F$.

First we have a K-isomorphism $\sigma_0 : K(\alpha) \to K(\beta)$. Using the isomorphism extension theorem, we can extend σ to a K-embedding $\sigma : F \to \bar{K}$. By our assumption, $\sigma(F) = F$. In particular, $\beta = \sigma(\alpha) \in F$.

$$\begin{tikzcd}
F \arrow[r, "\sigma"] & \bar{K} \\
K(\alpha) \arrow[u] \arrow[r, "\sigma_0"] & K(\beta) \arrow[u] \\
K \arrow[u] \arrow[r] & K \arrow[u]
\end{tikzcd}$$

$\square$

Lemma. Let F/K be an algebraic extension, and let $\sigma : F \to F$ be a K-embedding. Then $\sigma(F) = F$.

Proof. Let $\alpha_0 \in F$ be an element. Set $\text{irr}(\alpha_0, K) = (t-\alpha_0)(t-\alpha_1)\dots(t-\alpha_r)(t-\beta_1)\dots(t-\beta_s) \in \bar{K}[t]$ where $\alpha_i \in F$ and $\beta_j \in \bar{K}\backslash F$. Then σ permutes $\{\alpha, \alpha_1, \dots, \alpha_r\}$ so that $\alpha_0 = \sigma(\alpha_i)$ for some $0 \leq i \leq r$. $\square$

Remark. Let F/K be a finite extension. Then F/K is normal if and only if

$$[F : K]_s = |\text{Aut}(F/K)|$$

Lecture 16 (November 6th)

Remark. Last class we have learned normal extensions. This class, we learn more about normal extensions and newly finite fields.

Theorem. Let F/K be an algebraic extension and $\bar{K}$ be an algebraic closure of K with $F \subset \bar{K}$. The following are equivalent:

(i) There exists $S \subset K[t]$ such that F is a splitting field for S (that is, $F = K(S_0)$ where S_0 is the set of zeros of $f(t) \in S$).

(ii) For each K-embedding $\sigma : F \to \bar{K}$, $\sigma(F) = F$.

(iii) F/K is normal. That is, for each $\alpha \in F$, all zeros of $\text{irr}(\alpha, K)$ in $\bar{K}$ lie in F.

Example. (14.13) Consider $p(t) = t^4+t^3+t^2+t+1 \in \boldsymbol{Q}[t]$. Let $E = \boldsymbol{Q}(\xi)$, ξ is a primative 5th root of unity. Observe that σ_i can be defined to send any $\xi \mapsto \xi^i$ for $1 \leq i \leq 4$.

$$
\begin{tikzcd}
 & \bar{\boldsymbol{Q}} \\
E = \boldsymbol{Q}(\xi) \arrow[r, "\sigma_i"] & \boldsymbol{Q}(\xi^i) = \boldsymbol{Q}(\xi) \arrow[u] \\
\boldsymbol{Q} \arrow[u] \arrow[r] & \boldsymbol{Q} \arrow[u]
\end{tikzcd}
$$

Now consider $p(t) = t^3 - 2 \in \boldsymbol{Q}[t]$. Let $E = \boldsymbol{Q}(2^{1/3}, \xi)$ where ξ is a primative 3rd root of unity.

$$
\begin{tikzcd}
F = \boldsymbol{Q}(2^{1/3}) \arrow[r, "\sigma"] & \bar{\boldsymbol{Q}} \\
\boldsymbol{Q} \arrow[u] \arrow[r] & \boldsymbol{Q} \arrow[u]
\end{tikzcd}
$$

Here, $\sigma(2^{1/3}) = 2^{1/3}\xi \notin \boldsymbol{Q}(2^{1/3})$.

Corollary. Let F/K be a finite extension. Then F is normal over K if and only if

$$[F : K]_s = |\mathrm{Aut}(F/K)|$$

where $\mathrm{Aut}(F/K)$ is the group of all K-isomorphisms $\sigma : F \to F$, $\sigma \cdot \mu = \sigma \circ \mu$.

Remark. (Observation) Choose an algebraic closure of K with $K \subset F \subset \bar{K}$. There is an injection

$$\mathrm{Aut}(F/K) \to \{\sigma : F \to \bar{K}\}$$

where σ is a K-embedding. Schematically, we have a map

$$
\left(
\begin{tikzcd}
F \arrow[rr, "\sigma"] \arrow[dr, dash] & & F \arrow[dl, dash] \\
 & K &
\end{tikzcd}
\right)
\longmapsto
\left(
\begin{tikzcd}
F \arrow[r, "\sigma"] \arrow[dr, dash] & F \arrow[r, "\iota"] \arrow[d, dash] & \bar{K} \arrow[dl, dash] \\
 & K &
\end{tikzcd}
\right)
$$

Proposition. If F/K is a field extension of degree 2, then F/K is normal over K.

Proof. Choose an element $\alpha \in F \backslash K$. Then $F = K(\alpha)$. Let $\mathrm{irr}(\alpha, K) = t^2 + at + b \in K[t]$. Note that

$$t^2 + at + b = (t - \alpha)(t + (a + \alpha)) \in \bar{K}[t]$$

We claim that F is a splitting field for $\operatorname{irr}(\alpha, K)$. This follows from the observation that $-(a+\alpha)$ and α belong in F. An observation is that for $F/E/K$, if F/K is normal, then F/E is normal. $\square$